

Assembly Instructions for:

Revolver-II: single & double action

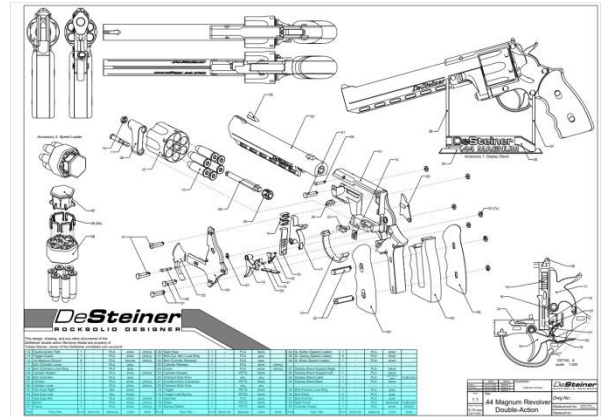
Rev.2 2024-09-15



Information

Part-Names

If you are unsure about the part names, you can check the explosion diagram. There is a full part list and by following the Part-Number of each part in the drawing, you will also find out what each part looks like and where it goes.



Printing instructions

I generally print rather slowly, as slower printing speeds tend to produce better results and more accurate models. This model has rather tight tolerances when it comes to the gaps between moving parts or any parts fitting together in general. In the end, I want the final print to work well, while also being pretty stable. If I loosened the tolerances by increasing the gaps between the moving parts or parts that are supposed to be clamped together, the printed model would flex around like crazy. That would suck, right? Also, if a fit is too tight you can always grind away a little bit of material, but you cannot simply add material when a fit is too loose.

So here is my typical setup:

- **layer height:** 0,15mm
- **outer perimeters:** 20mm/s
- **other perimeters:** 25mm/s
- **infill:** 30mm/s
- **max. acceleration:** 300mm/s² (to prevent tall models from wobbling around)

None of the models require supports. Because I went the extra mile to design everything especially with FDM-printing in mind, so you will save some filament otherwise wasted on support material, as well as the time it takes to remove that stuff. You are welcome ;)

I use brims on all of my prints to really make sure that they stick nicely to the heatbed. Some models may have features included that improve brim generation and are intended to be removed before assembly. One alternative to brims, that I have used more often recently, is to add Cylinders to the model in the slicer, which are just one layer high and ~3mm in diameter, in any spots that are endangered to lift from the heatbed, like any outside corners and stuff. This provides even more stability than a brim and is much less effort to remove.

All of the parts are designed with flat surfaces for good first layer adhesion and they are all oriented correctly right out of the box. I also include small chamfers around these first-layer surfaces to counteract the elephant-foot-syndrome. However, they make the parts more vulnerable towards part-lifting due to warpage, which is why I recommend brims.

If you own a Prusa and want the *Display Stand Label* or the *Sight Front* to be extra pretty, you can make use of the M600 filament change gcode (which is directly included into the PrusaSlicer as of now). Otherwise you have to wait for the right layer height to come up and initiate the color change manually, by simply pausing the print and changing the filament.

Before assembly

Make sure all parts are completely free of strings, blobs, first layer burrs (elephant foot) and so on. Like already mentioned above, tolerances are tight.

Lubrication

To promote smooth operation of the Revolver, it is good to apply lubrication on critical surfaces. These are mentioned throughout the assembly steps, by an oil-lamp symbol you probably recognize from cars.

Lubrication will reduce friction between moving parts and will allow the Revolver to work as good as possible.

Either you do this after you assembled it once to check if it all fits together, or directly during the first assembly. That is up to you.

You may even use butter as lube (highly professional, I know ^^) as its biological, not too fluid and easily available. Seriously, no need for high-performance lubes here. Anything will do, as long as it is fluid enough but not too fluid that it creeps all over your model. Eventually just touching the critical surfaces a lot without any lube will already be sufficient, as you spread the small amounts of fat, which is naturally generated by your skin, on the model.

Only valid source

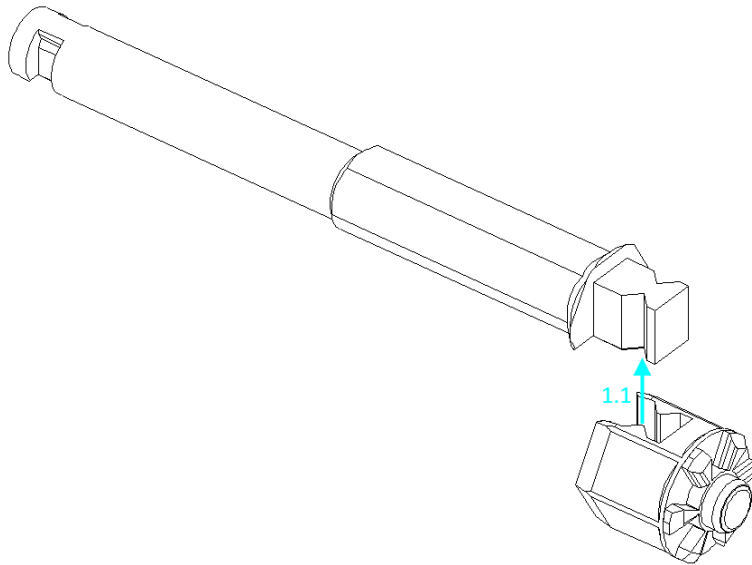
Make sure you downloaded the files from the “DeSteiner” account on printables.com. If there is any other source, someone stole my creation and is likely making money by offering it. This can be very frustrating. I have seen this in the past with my first Revolver design multiple times. I offered those files for free, someone uploaded them again on another page and charged 130\$ (!!!) for it! That’s absolutely hilarious. Two of those stolen downloads are worth more than what I got from donations over 5 years, can you imagine that?!

If you happen to find other sources for my files, please reach out to me and maybe also directly hit up those idiots.

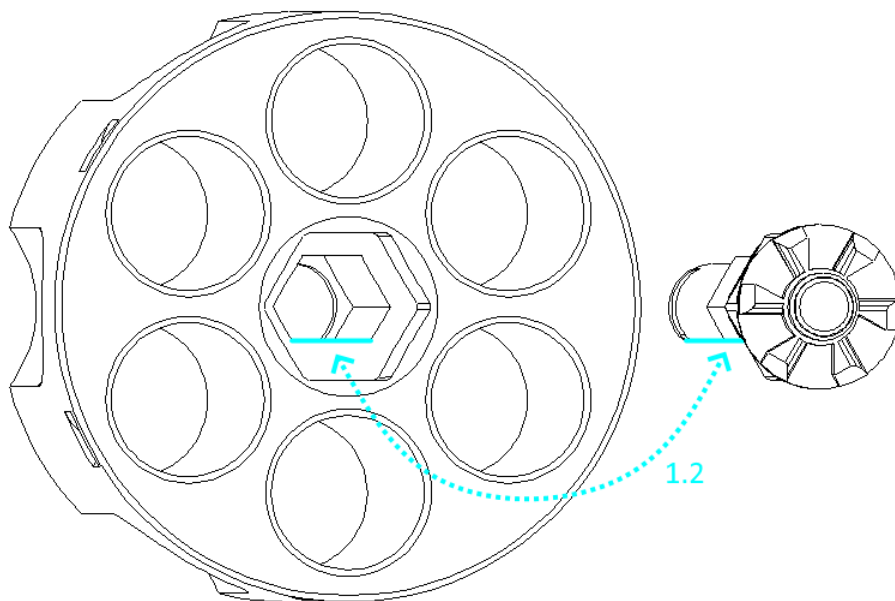
Problems?

If you run into any kind of troubles with this model, feel free to ask me. If you are friendly and properly describe your issue, I am happy to help! Sometimes the models print very good for me but others run into issues, or I overlooked something. Whenever you point out an error and I fix it, countless other guys will save some time as they don’t run into the same problem again.

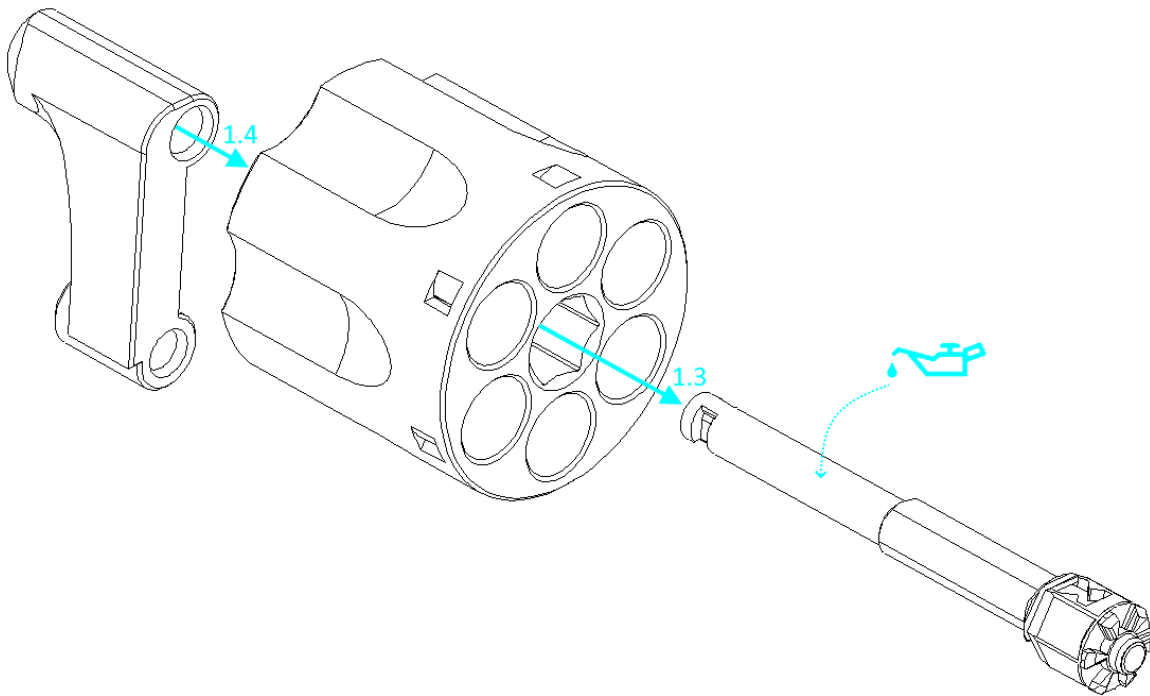
Step 1: Cylinder assembly



- 1.1** Take part *08 Bolt (Cylinder)* and slide *09 Cylinder Rotator* onto the dovetail from the side where the *Bolt* is flat.

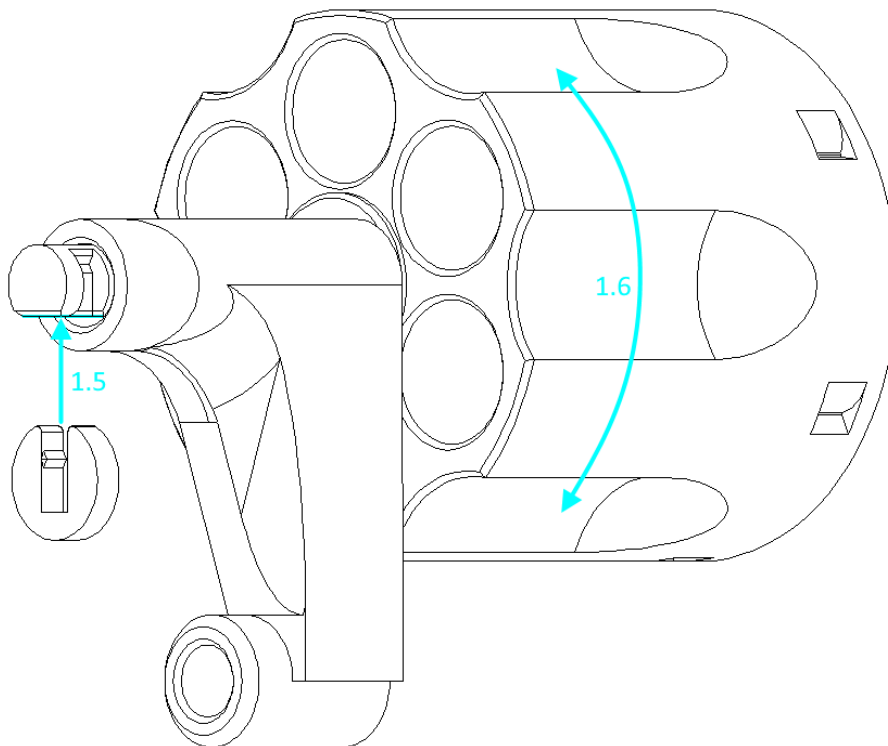


- 1.2** For the next step, note that the hole in *07 Cylinder* is not perfectly round/hexagonal, one side is flat, fitting to the flat surface of the *Bolt*. They must be aligned for assembly.



1.3 Push the *Bolt* into *07 Cylinder*. Lube the round surface of the *Bolt*.

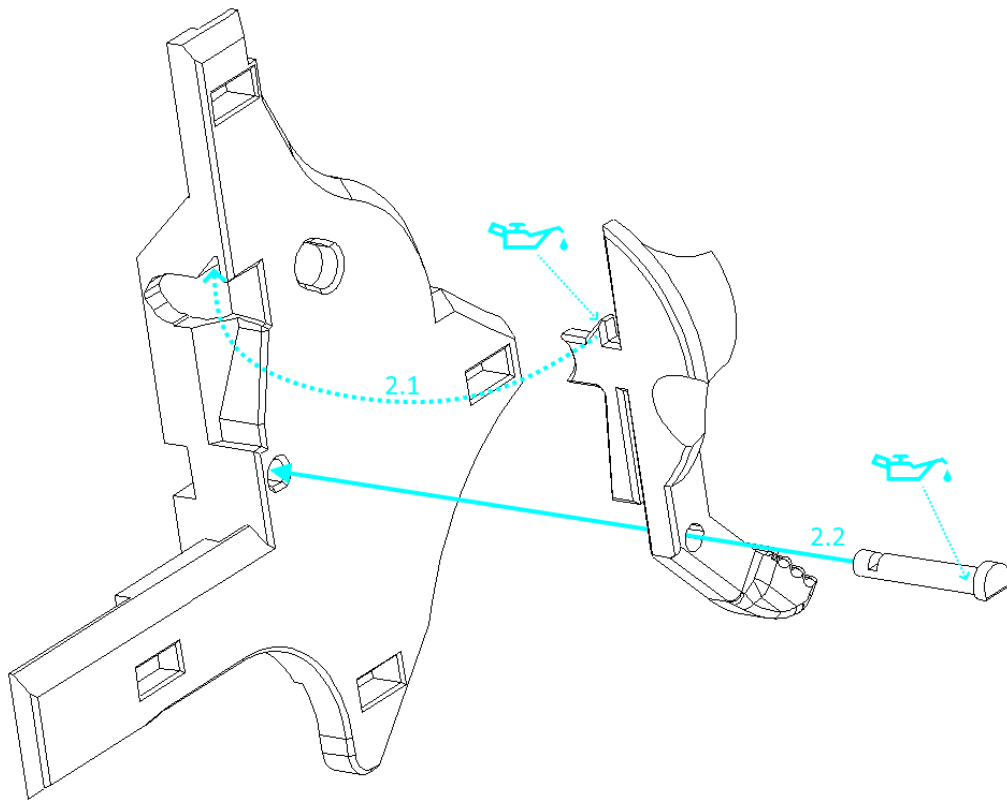
1.4 Slide *06 Cylinder Lever* onto the round surface of the *Bolt*.



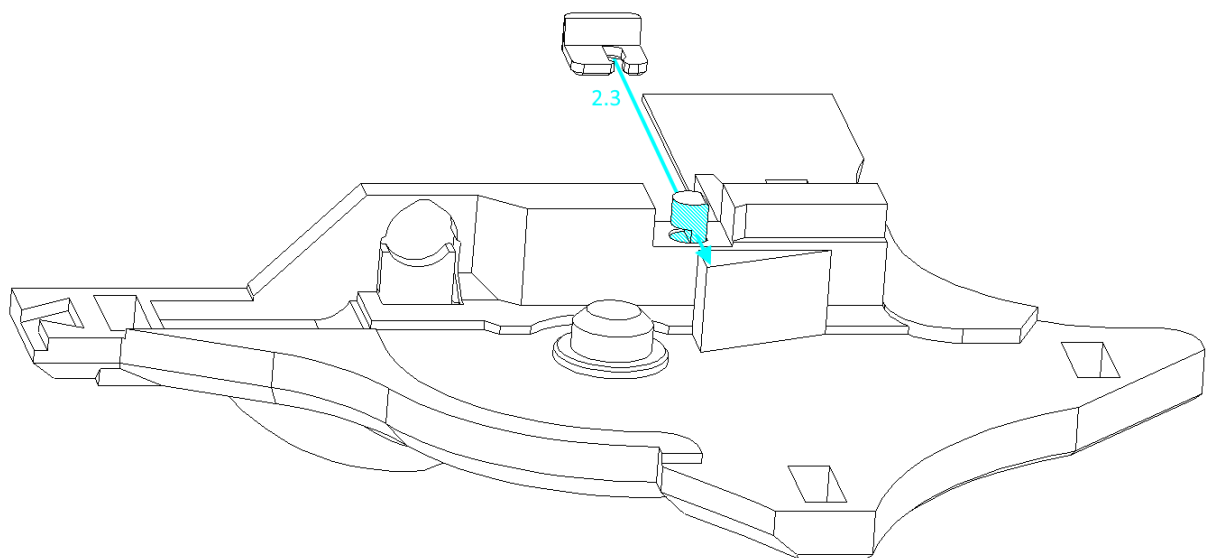
1.5 Push the *10 Bolt (Cylinder) Lock Ring* onto the *Bolt*, from the direction where the *Bolt* is flat.

1.6 Spin the *Cylinder* a few times. If it stops immediately after you let go, there is too much friction. Take the *Lever* off and remove any bottom/top layer burrs or excess material of the *Bolt* until the *Cylinder* keeps spinning for a short time.

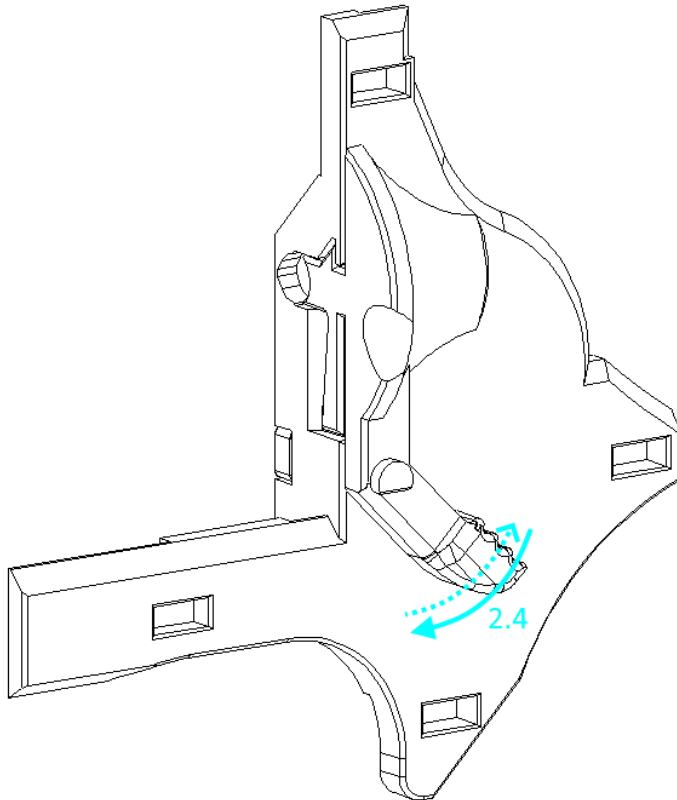
Step 2: Cover assembly



- 2.1** Put 25 *Cylinder Release* onto 24 *Cover* with the triangular notch of the *Release* inside the corresponding triangular recess in the *Cover*.
- 2.2** Mount the *Release* onto the *Cover* by pushing 26 *Bolt (Cylinder Release)* into the *Cover*. Again, the *Bolt* has a flat surface and the hole in the *Cover* is not round, so the flat surfaces must be aligned. Apply lube to the round surface of the *Bolt* and to the triangular notch of the *Release* for smooth operation.

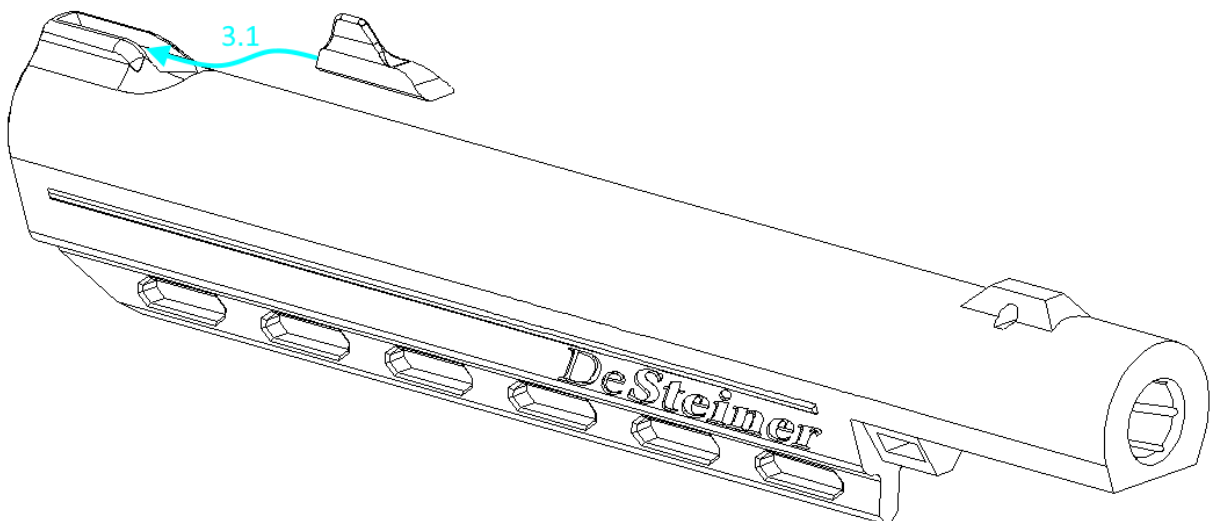


- 2.3** Lock the *Bolt* in place by pushing 27 *Bolt (Cyl. Rel.) Lock Ring* onto the *Bolt*.



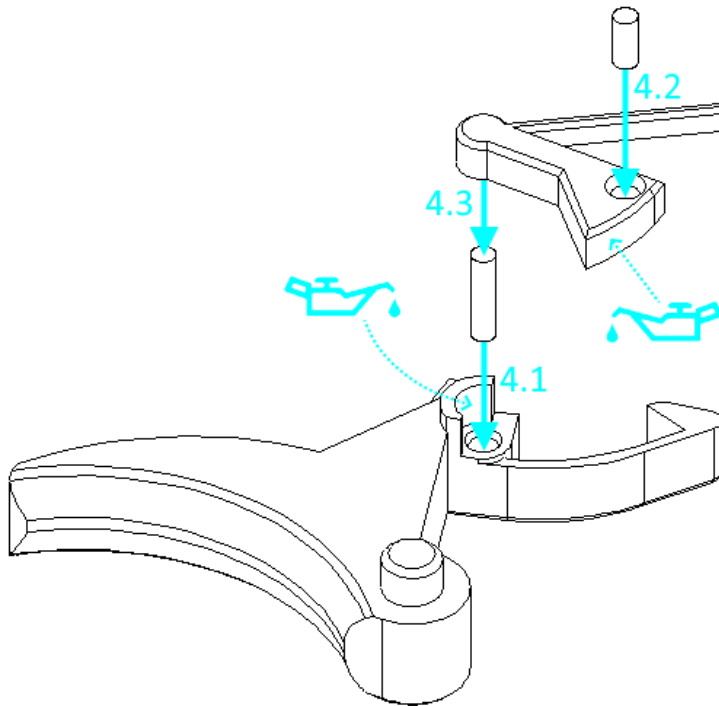
- 2.4** Push the *Release* a few times. It should jump back into the original position each time. If this is not the case, disassemble again and remove burrs/excess material from the *Bolt* or from the triangular notch of the *Release* until the *Release* springs back as intended. Holding only the *Bolt* while spinning the *Release* on it a few dozen times can also help.

Step 3: Barrel assembly



- 3.1** Push *30 Sight Front* into the cutout of *02 Barrel*. This is supposed to be a press fit, so some force will be necessary. Only one of the *Barrel* variants must be printed, they are all installed equally.

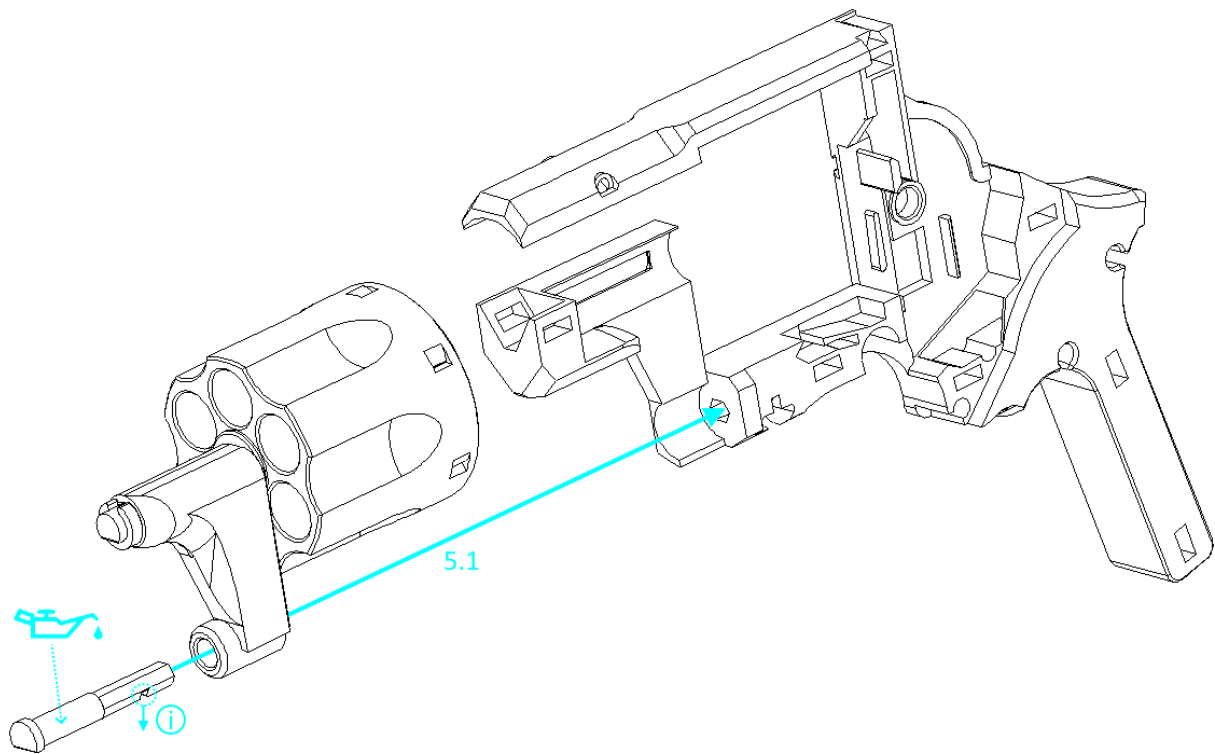
Step 4: Trigger assembly



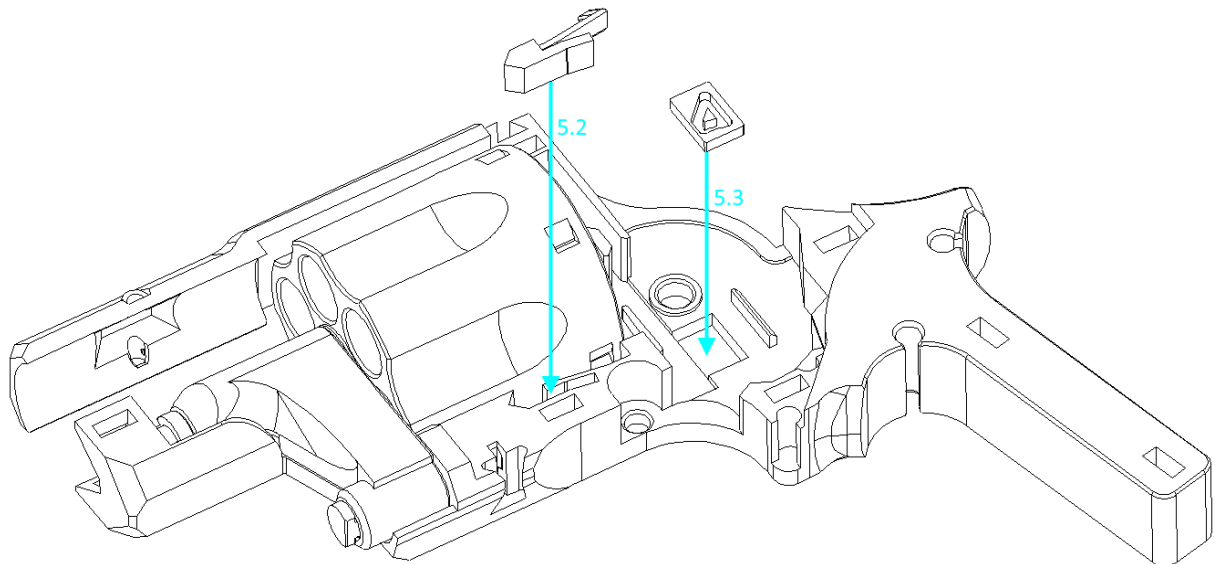
- 4.1** Push 20 *Filament Stub 7mm* into the hole of 19 *Trigger*. Lube the inner round surface of the *Trigger* around this filament “axle”.
- 4.2** Push 22 *Filament Stub 4mm* into the hole of 21 *Double-Action Connector*. Lube the top round surface of the *Connector*.
- 4.3** Put the *Connector* onto the filament stub inside the *Trigger*.

All sub-assemblies are prepared, time to build the Revolver!

Step 5: Revolver assembly

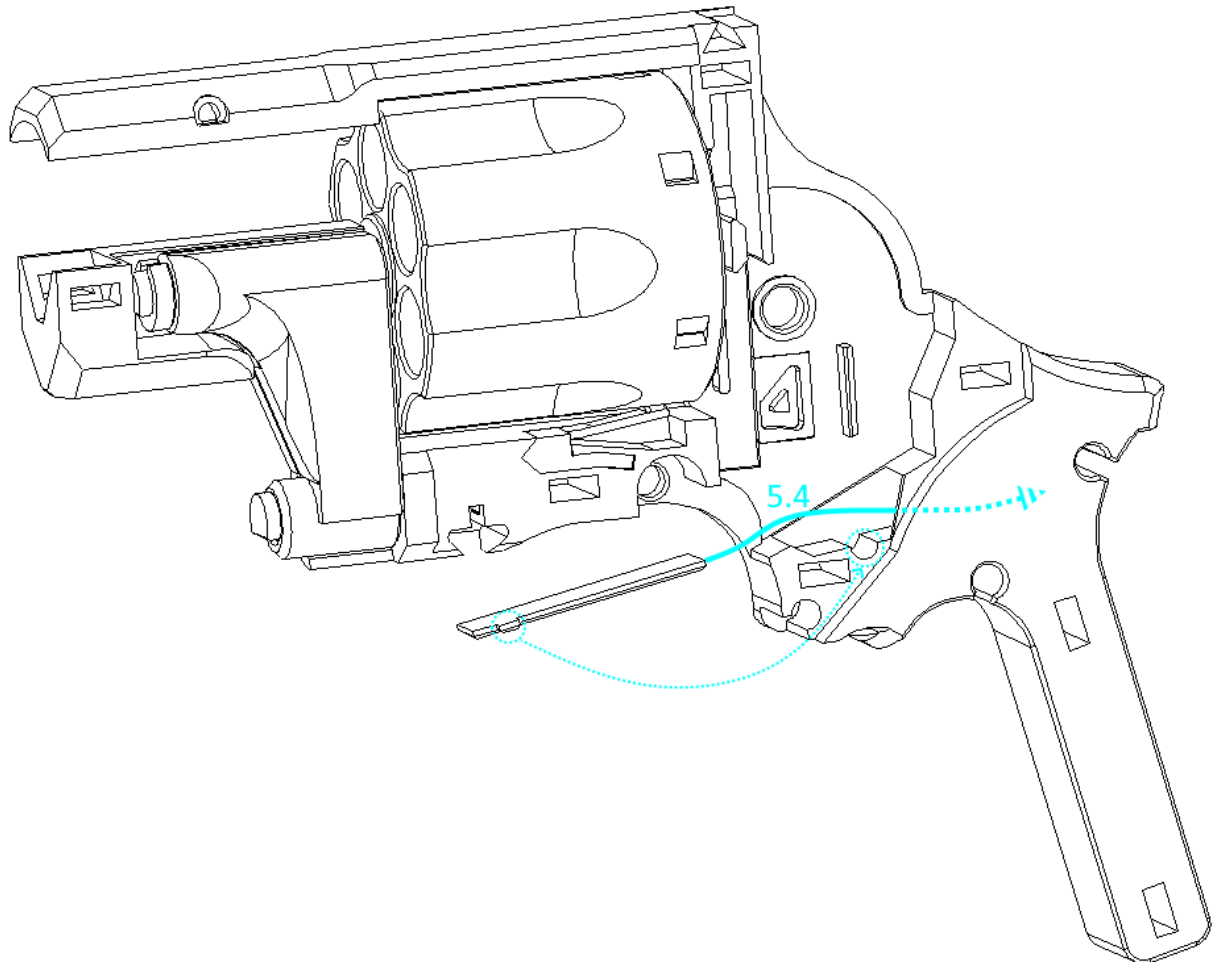


- 5.1** Mount the *Cylinder* sub-assembly onto *01 Frame* by pushing *11 Bolt (Cylinder Lever)* through the *Lever* into the hexagonal hole of the *Frame*. Lube the round surface of the *Bolt* so that the *Cylinder* can flip in and out easily. The cutout and flat surface of the *Bolt* must face downwards as shown.

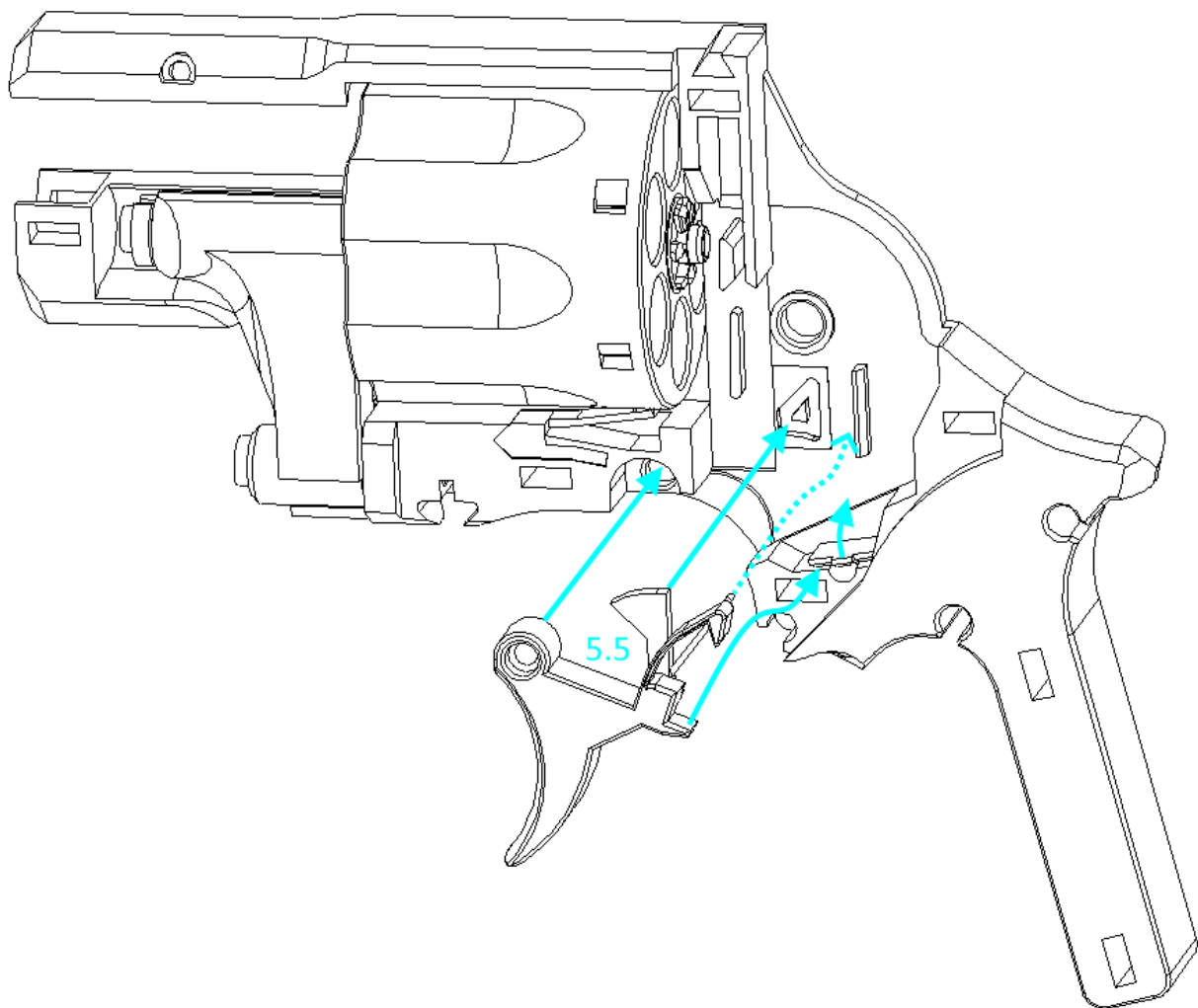


- 5.2** Push *23 Cylinder Stopper* into the *Frame*.
- 5.3** Drop *14 Double-Action Path* into the *Frame* with the shown orientation.

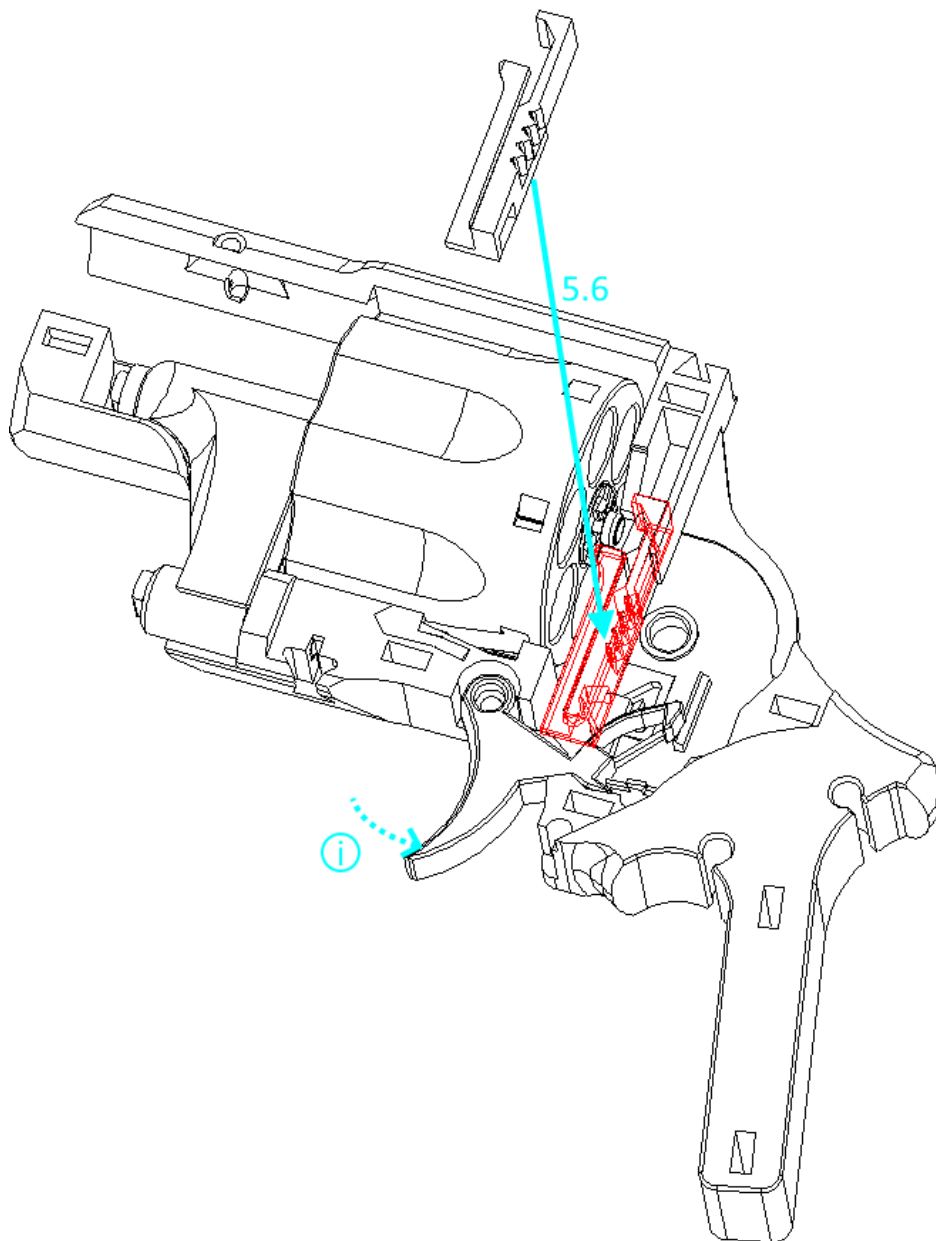
If single-action (manually cocking the *Hammer*, then pulling the *Trigger*) does not work properly anymore, e.g. the *Trigger* releases the *Hammer* and pulling the *Trigger* further rotates the *Cylinder* once more and the *Hammer* fires again, then your *Double-Action Path* is probably broken. To avoid scrapping the entire frame, this geometry is an easily replaceable separate part.



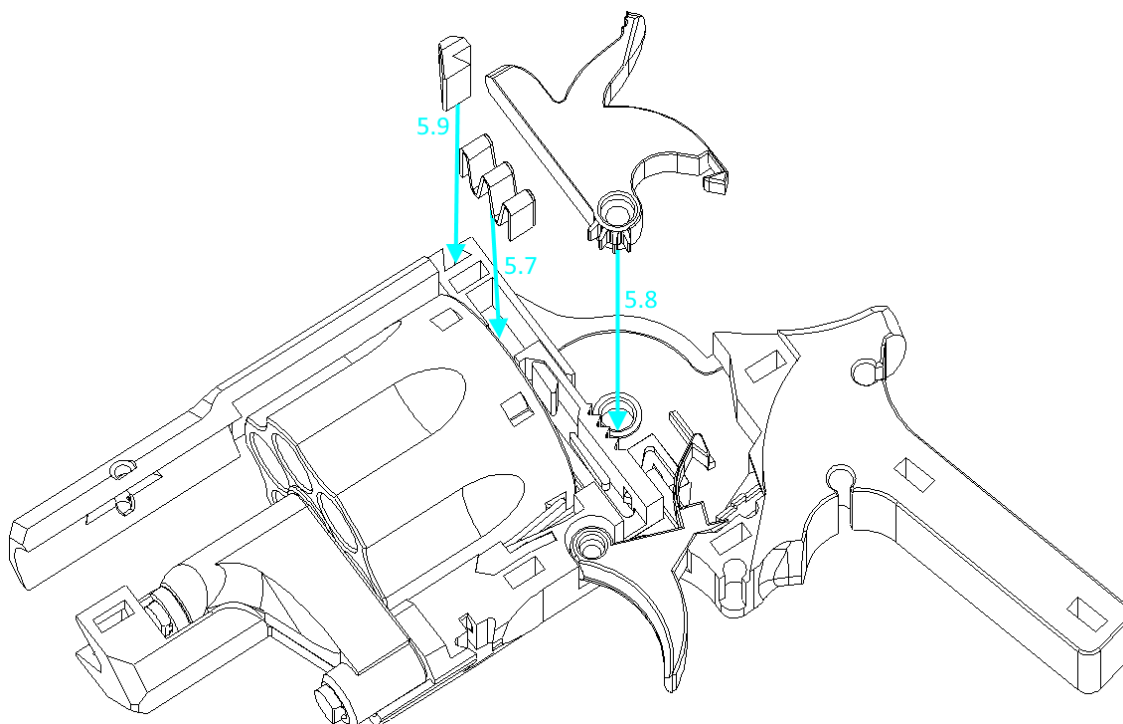
- 5.4** Slide 18 *Trigger Leaf-Spring* into the slit inside the *Frame*. Fully inserted, the notch on the side of the *Leaf-Spring* should be aligned with the half-circle cutout of the *Frame*.



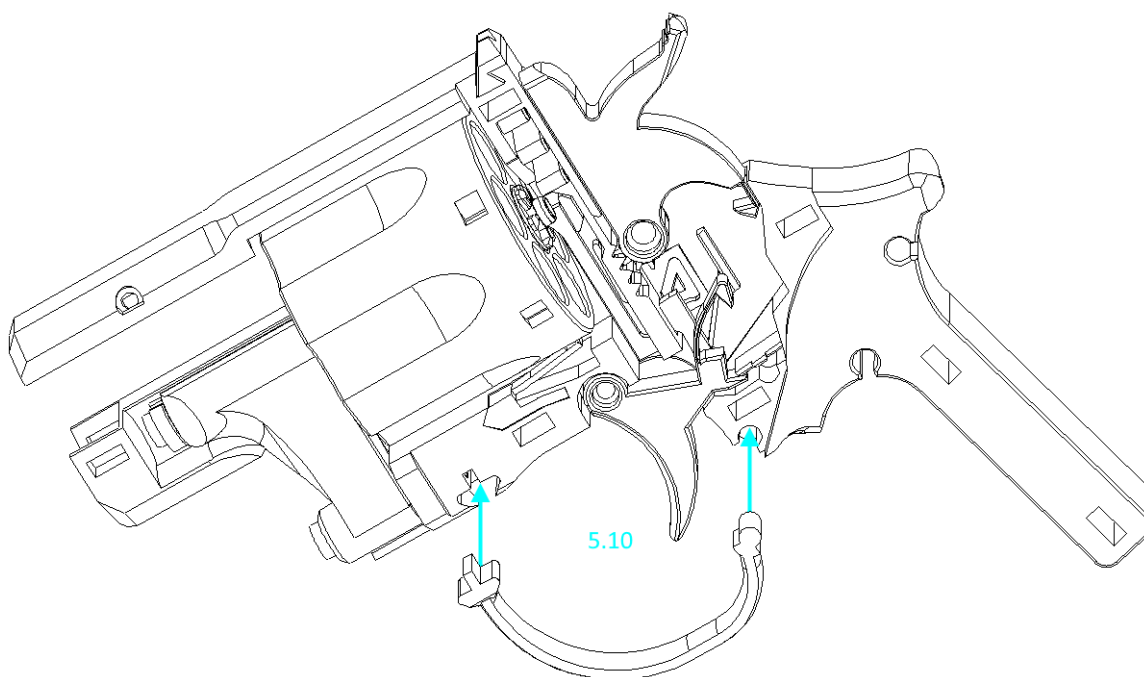
- 5.5** Install the *Trigger sub-assembly* in the *Frame*. Note that:
- 1) the rear notch of the *Trigger* must be below the *Leaf-Spring*.
 - 2) the *4mm Filament Stub* of the *Double-Action Connector* must be inside the groove of the *Double-Action Path*.
 - 3) the small leaf-spring of the *Double-Action Connector* must press against the small rib of the *Frame*.



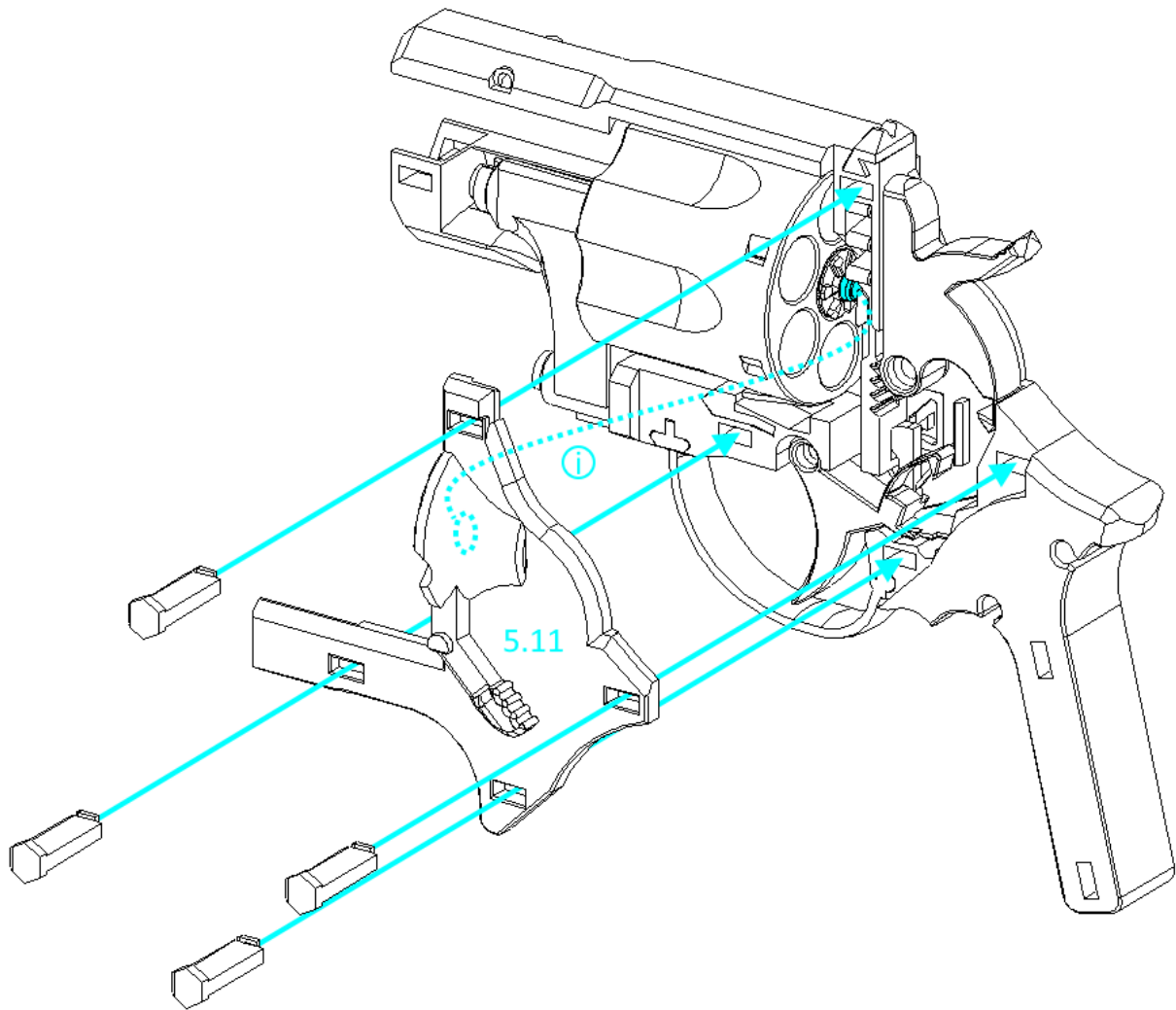
- 5.6** Place 16 *Slider* in the *Frame*. The *Slider* has a cutout where the *Double-Action Connector* goes into. Pulling the *Trigger* (while also holding it down) will move the *Connector* up and back so that the *Slider* can be inserted easily.



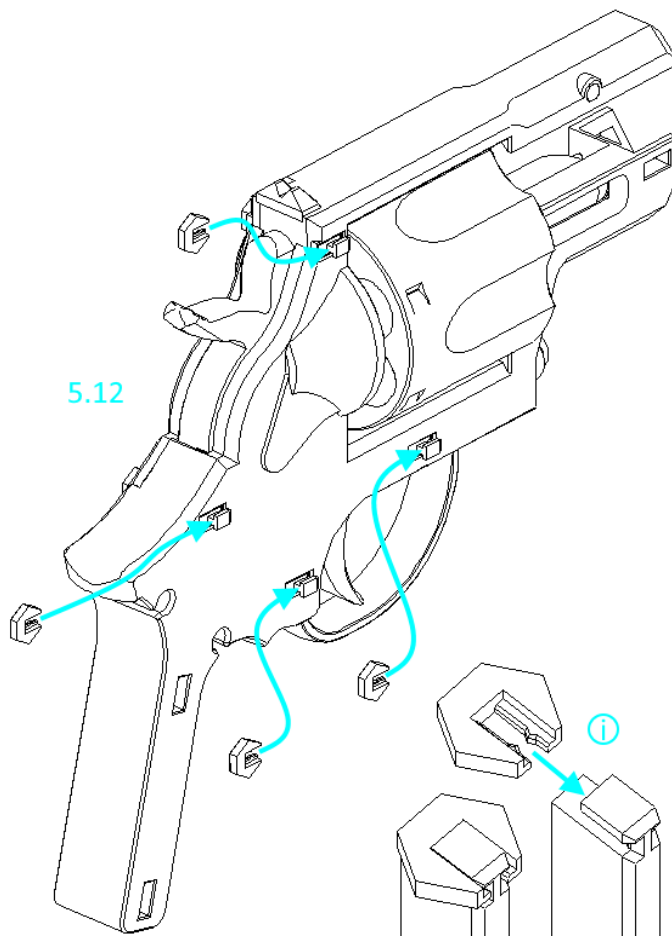
- 5.7** Place *15 Spring (Slider)* in the rectangular space above the *Slider*.
If it happens later that the *Hammer* does not return all the way to its original position after pulling the *Trigger*, it is likely that the *Spring* broke down and needs to be replaced. Bad seam alignment along the bends or resting in the cocked position for a long time can lead to early *Spring* failures.
- 5.8** Place *17 Hammer* in neutral position in the *Frame*.
- 5.9** Install *28 Sight Rear* at the dovetail at the top of the *Frame*.



- 5.10** Push in *13 Trigger Guard* until it is level with the surface of the *Frame* around it.



- 5.11** Seal the internals with the *Cover sub-assembly*. The *Cover* with the *Cylinder Release* functions as a bearing for the *Cylinder*. Therefore it is necessary to flip the *Cylinder* out to the side a little bit, aligning the *Covers* bearing spot with the knob of the *Cylinder*, and finally pushing the *Cover* fully onto the *Frame*. Secure the *Cover* with 4 pieces of *31 Bolt (Frame)*.



5.12 Lock the 4 *Bolts* in place with 4 pieces of 33 Bolt (Frame) Lock Ring. Push them onto the *Bolts* as shown.

Story-time!

Are free models, like this one, a feasible source of income?

Sounds legit. After all, my models are high quality and fairly popular, right?

Well, here's the data: after **6 years** I got **550.000 downloads** on all my models, invested probably **500h of effort**, and got **330\$ worth of donations...**

...That is less than 0,001\$ per download.

...Less than 1\$ per hour of work.

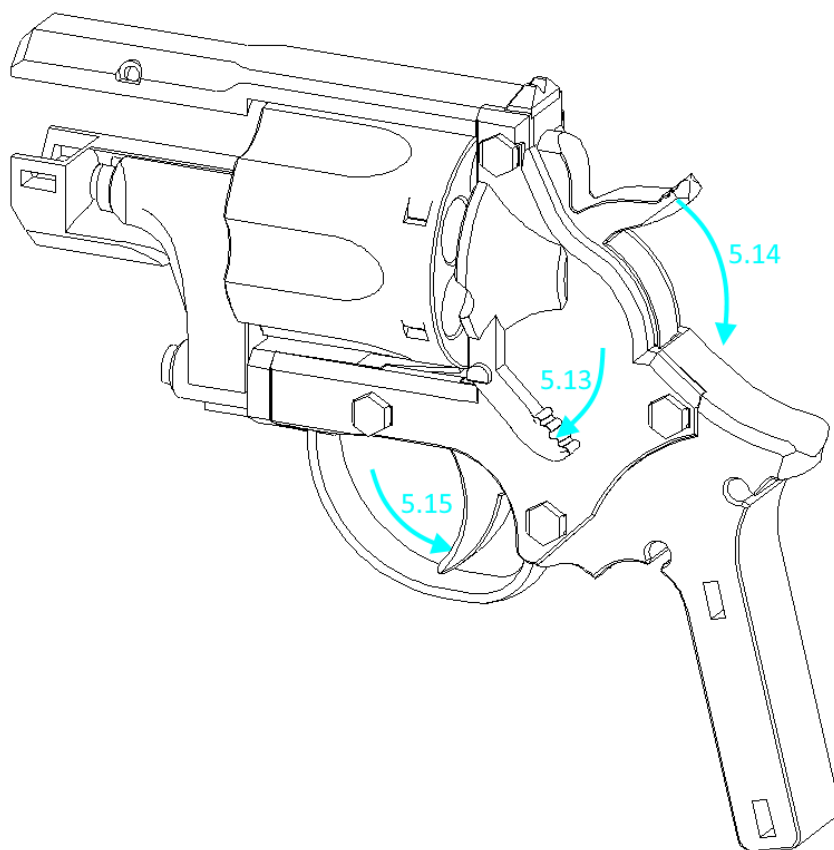
Not very rewarding, is it?

Ask yourself: Would you spend another 100h on the next project under these conditions?

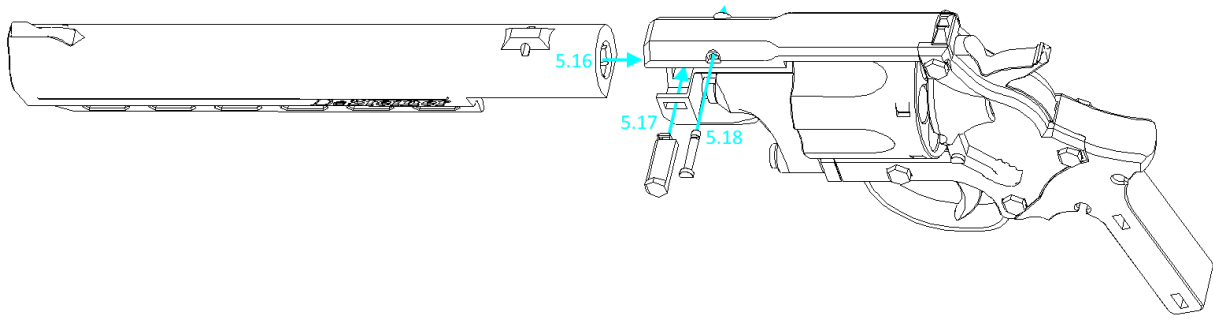
Honestly, sometimes I feel like an idiot doing all of that for free... while others are pricing their models of similar quality between 20\$ and 40\$ per download.

Now it's up to you. Do you want me to keep going? Do you want to see more free stuff made by me? Then show me that my effort IS in fact worth it!

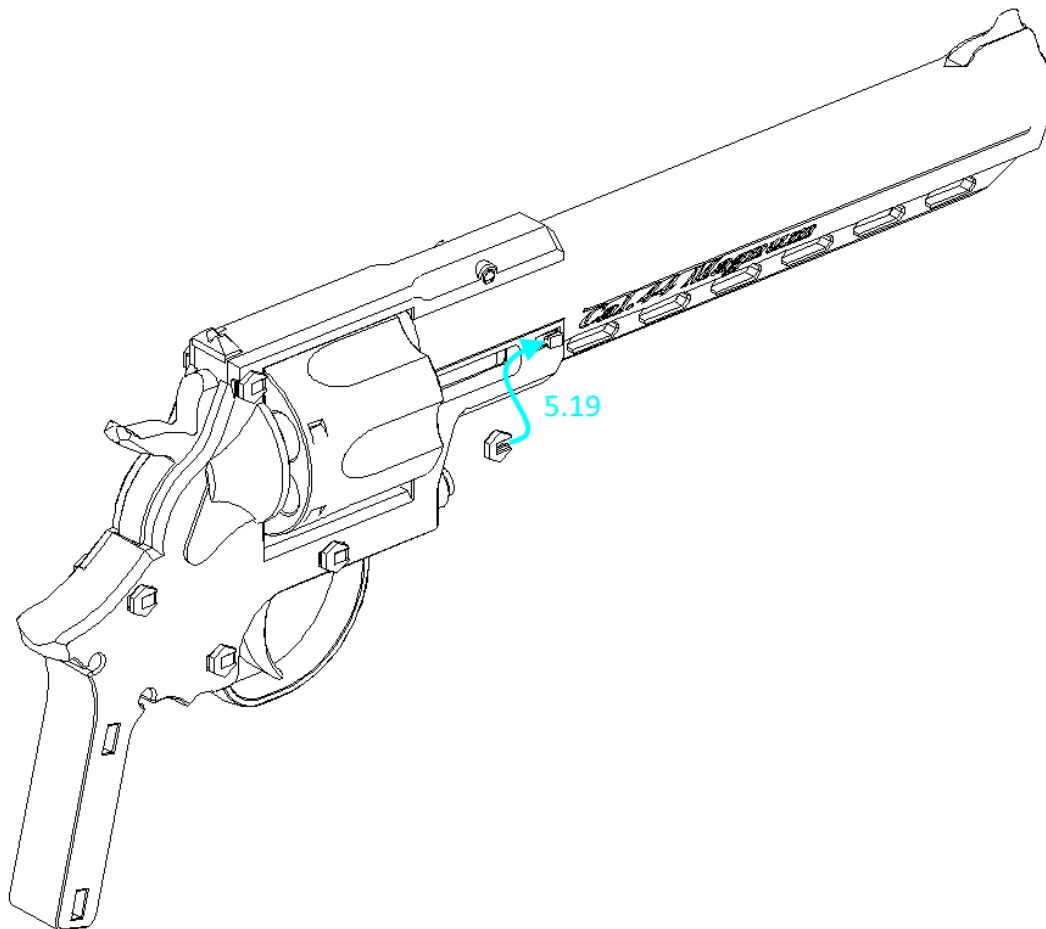
If you have paypal, just follow www.paypal.com/paypalme/tosteiner and donate whatever amount you think is appropriate. Receiving money for free things is the ultimate sign of appreciation, and it really makes my day, every single time! So what are you waiting for? ;)



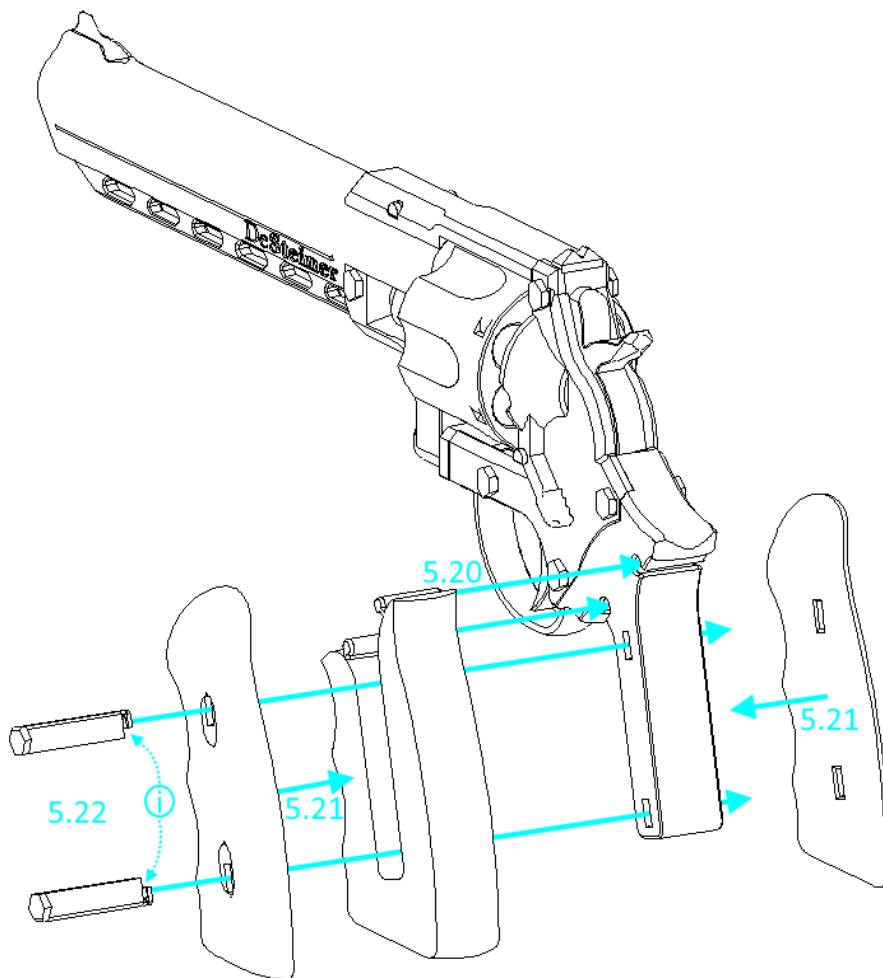
- 5.13** Time for a functional check. Start by pressing the *Cylinder Release*. Twisting the *Revolver* to the side, best case would be that gravity is enough to flip the *Cylinder* out of the *Frame*. At **least** a whip-movement should be sufficient to flip the *Cylinder* out. If this is not the case, find spots with excess material and remove until it works. Then with the *Cylinder* out, let go of the *Release* and try to whip the *Cylinder* back in again. It should flip fully into its neutral state with the *Release* closing again.
- 5.14** Next, try the single-action by pulling the *Hammer*. The *Cylinder* should rotate and snap in place in the next of its 6 notches. Pulling the *Trigger* should bring the *Hammer* all the way to neutral state, and letting go of the *Trigger* should also bring it back to neutral state on its own. Note that pulling the *Hammer* too fast may rotate the *Cylinder* too far, as the *Stopper* can only withstand a certain amount of inertia of the *Cylinder*.
- 5.15** Finally, test the double action by gently pulling the *Trigger* without cocking the *Hammer* first. Again, the *Cylinder* should rotate and snap into place after 60° while the *Hammer* moves backwards before releasing and ‘firing’. Letting go of the *Trigger* must bring it back to neutral state on its own. If any of that does not work properly or the whole operation is not smooth (it requires some force, but should not be too much) then you have too much friction in your system and need to optimize with sanding and/or lubing. Again, pulling the *Trigger* too quickly will probably rotate the *Cylinder* too much.



- 5.16** Slide the *Barrel sub-assembly* into the *Frame*.
- 5.17** Attach the *Barrel* to the bottom part of the *Frame* by pushing a *31 Bolt (Frame)* through the rectangular hole.
- 5.18** Attach the *Barrel* to the top part of the *Frame* by pushing a *26 Bolt (Cylinder Release)* into the *Frame*. As the *Bolt* is not fully round, correct orientation is mandatory.



- 5.19** Lock the bottom *Bolt* in place with another *33 Bolt (Frame) Lock Ring*.

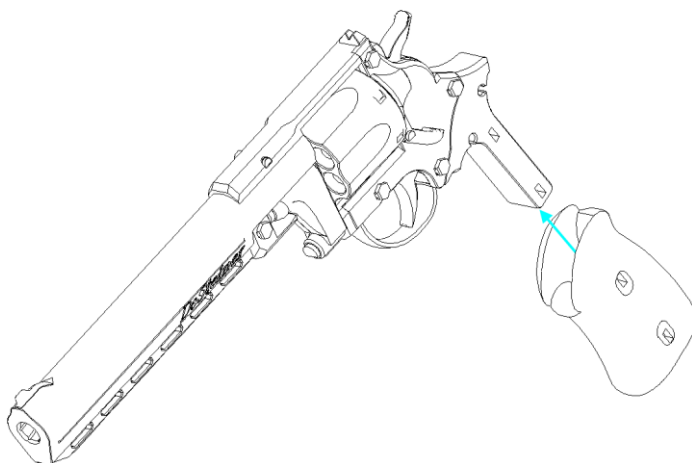


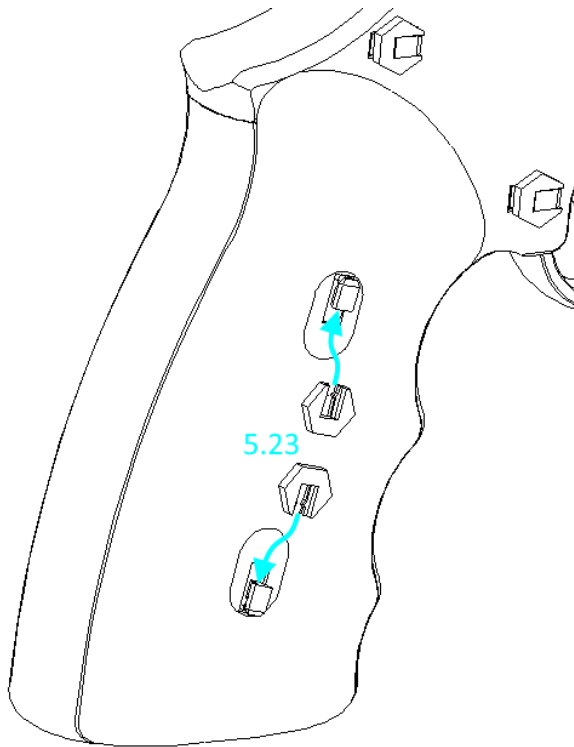
5.20 Slide *03 Grip Ergo Mid* onto the *Frame*.

5.21 Place *04 Grip Ergo Left* and *05 Grip Ergo Right* to the sides accordingly.

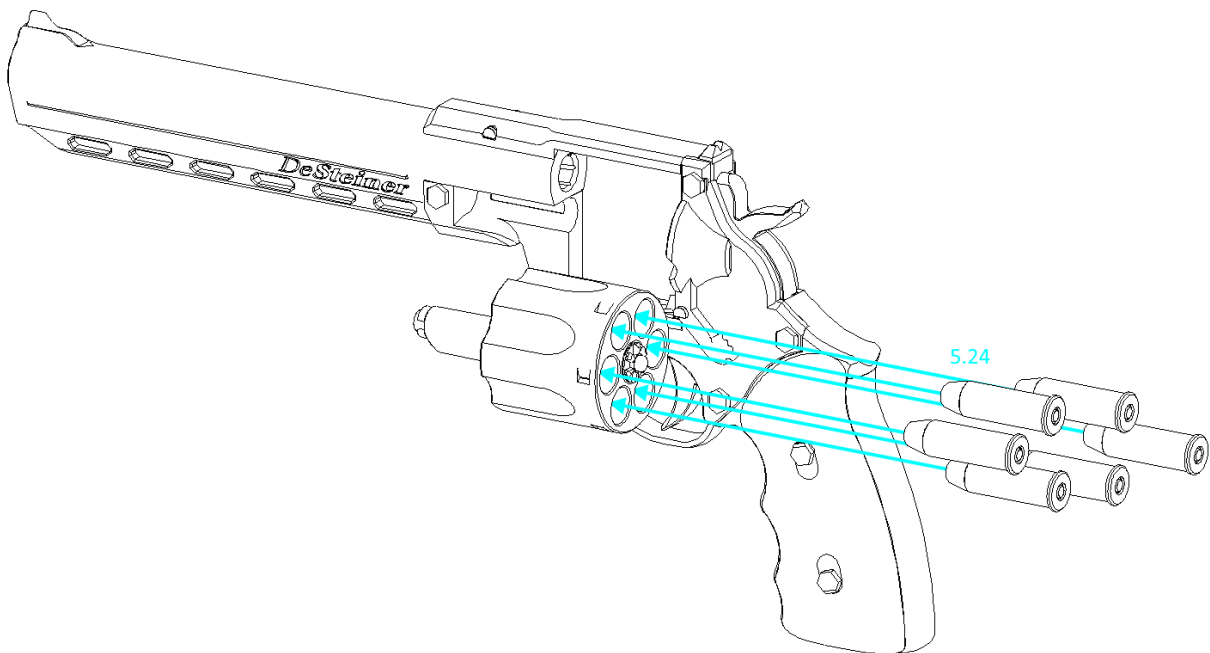
5.22 Push two *32 Bolt (Grip)* (longer than *Frame Bolts*) through the holes. Pay attention to the correct orientation, or you will not be able to install the *Lock-Rings* later on.

You only have to print either the *ergonomic mid+left+right*, the *classic mid+left+right*, the *ergonomic full* or the *classic full* parts. The *ergonomic* and *classic* versions are installed equally. The *full* versions are simply slid up the *Frame* and then locked with the same *Bolts*.
Tip: the left+right or the full grips feel great when printed with flex-filament!





5.23 Lock both *Bolts* in place with two more 33 Bolt (Frame) Lock Rings.

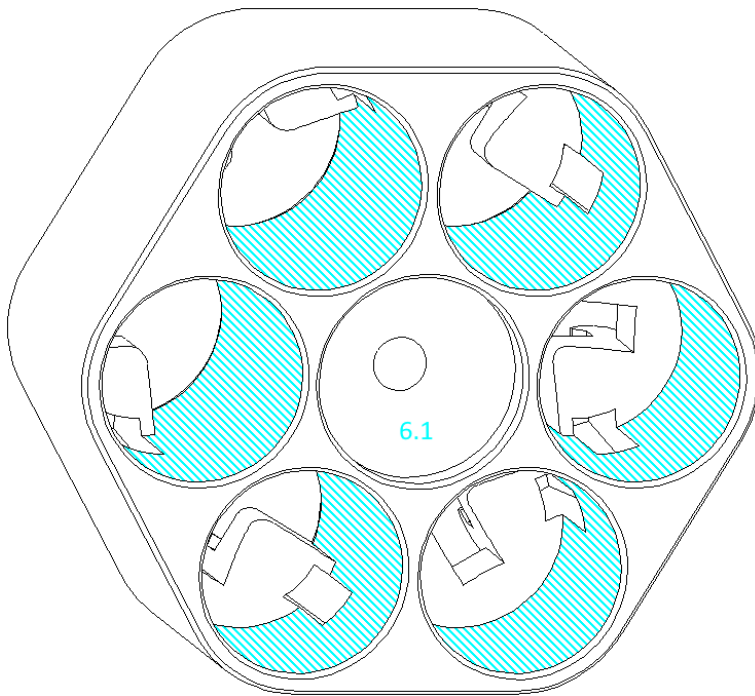


5.24 All done! Time to load up and have some fun with your new toy!

Unless you also printed the accessories, then there is still some more work to do ;)

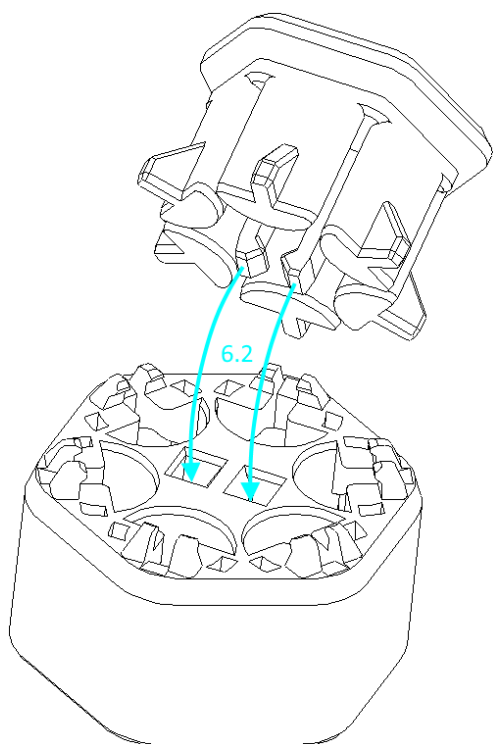
Quick check: have you made a donation yet? By now you probably already got more excitement out of the revolver than you got from the last overpriced coffee you bought. Think about it...

Step 6: Speed-Loader assembly

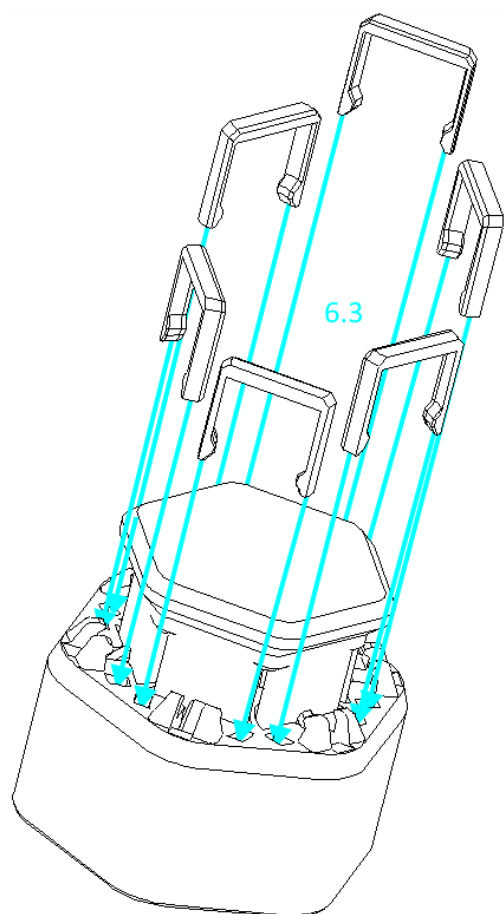


- 6.1** Take *38 SL Body* and make sure, that the holes for the bullets are free of blobs or strings and remove any bulges of seams as much as possible. One way of doing this, is to take one of the *Magnum Rounds* and move it up and down in each of the holes while pushing the rim against the hole walls. This will grind away the uneven spots.

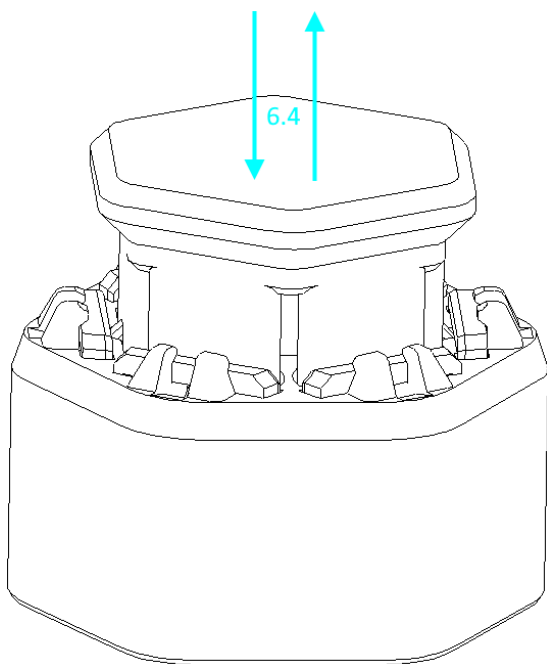
Tip: print the *SL Body* with a different layer-height than *SL Button*. This will make the *Button* push much smoother, as the layers are unable to align and will not get stuck in each other.



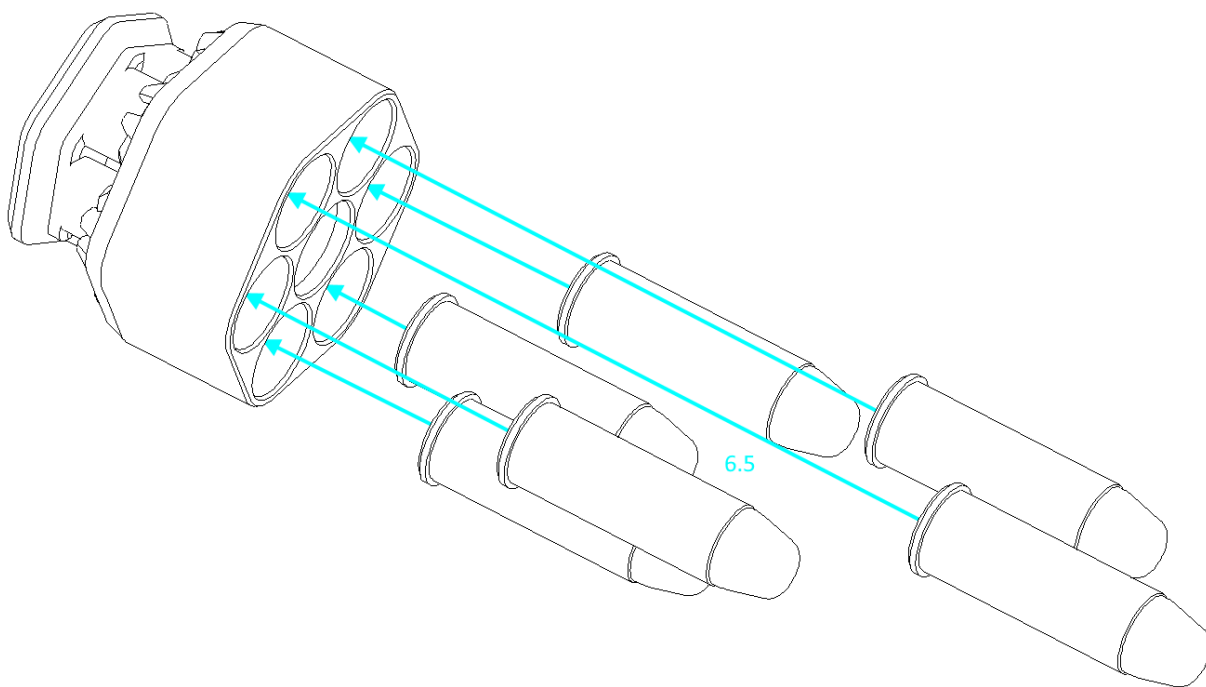
6.2 Install 40 SL Button by pushing it into the *Body*.



6.3 Insert the 6 pieces of 39 SL Spring at the cartridge holes of the *Body*. Push them over the noses of the *Body*, which prevent the *Springs* from falling back out again.

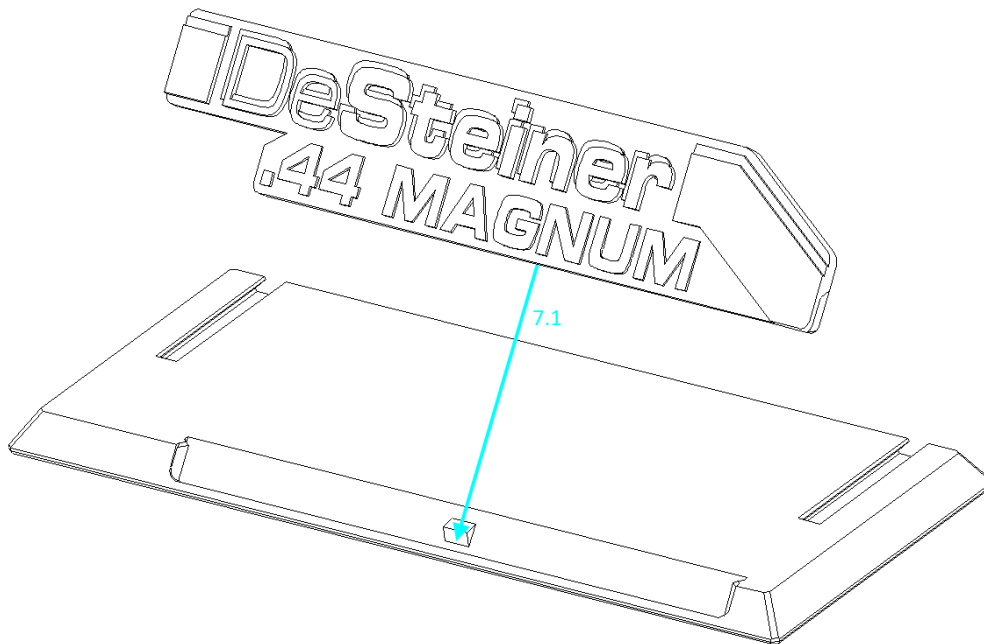


- 6.4** Push the *Button* in and out repeatedly (several dozen times) until the movement feels smooth, similar to step 6.1

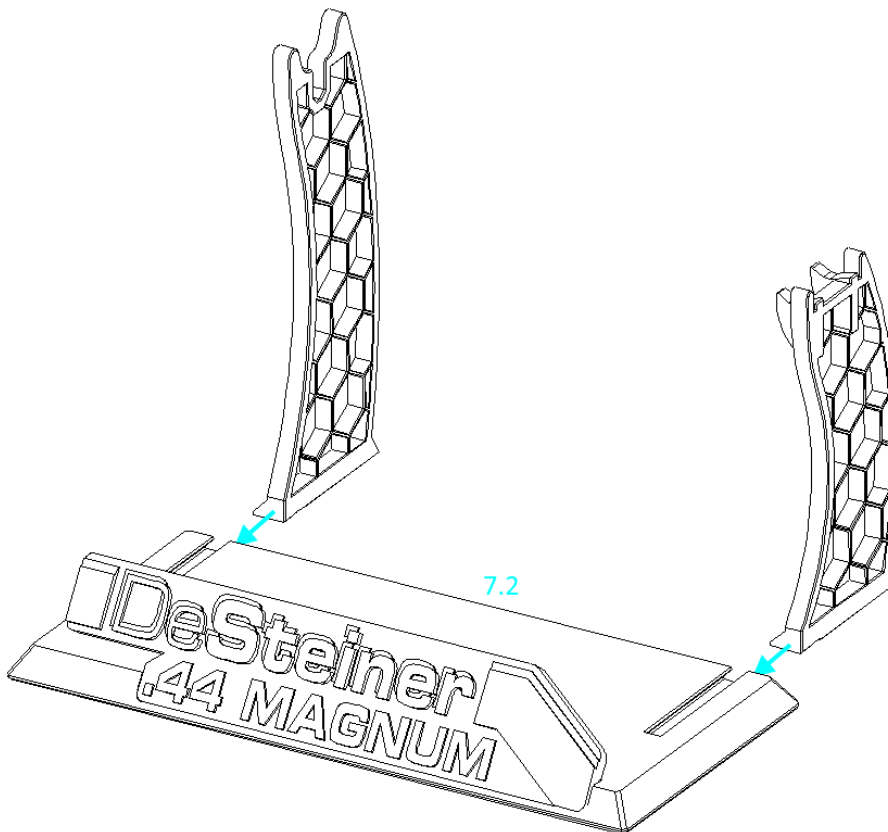


- 6.5** Insert the *Magnum Rounds* and you are done! They are held in place by the *Springs*, pushing the *Button* will release them into the *Cylinder* of the *Revolver*.

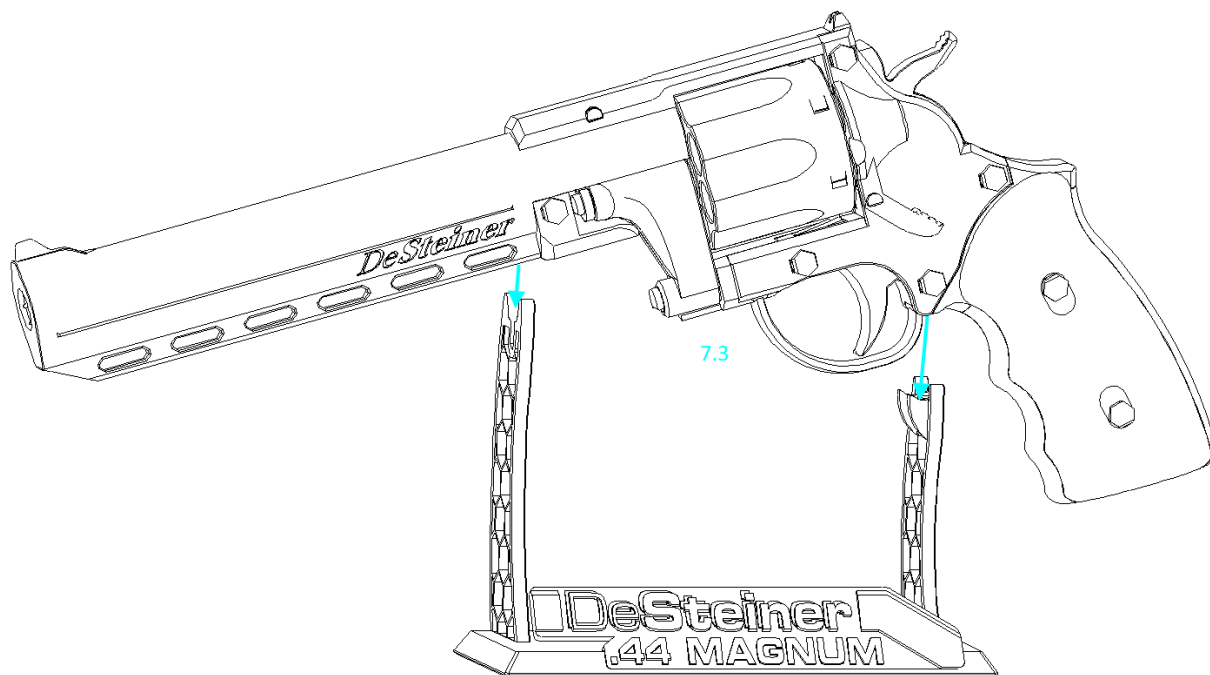
Step 7: Display-Stand assembly



- 7.1** Take 34 *Display-Stand Base* and install 35 *Display-Stand Label*. If you would like to leave my brand-name or the entire *Label* away, remind yourself that you got this high-quality model for free while others would probably charge 50 \$ for it, and reconsider your choices ;)



- 7.2** Slide the taller 36 *Display-Stand Support Left* into the left slit, and the lower 37 *Display-Stand Support Right* into the right slit of the *Base*.



7.3 The *Display Stand* is complete. Put the *Revolver* on the *Stand* so that the round edges next to the *Trigger* are resting on the two dedicated support surfaces of the right *Stand Support*, and admire your marvelous print! ;)

Tip: the hexagonal holes of the Supports have the perfect size to stick the Magnum rounds into, in case you have printed some extra and need a place to put them.

Congratulations! Assembly is complete!
Have fun playing around with your new toy!
And don't forget to share your makes, I'd love to see them.

Did you notice that you never needed glue or screws at any point? You can disassemble and reassemble this model at any given time, without any tools! Just like a 3d-puzzle.

Step 8: huge effort for free...

As you can imagine, a whole lot of time and effort was invested into this. Even though I already made a single-action Revolver in 2018, all parts were overhauled again, using the knowledge I gained in the meantime. Making such a design with 3d-printing in mind really increases the effort. Instead of simply designing, you have to plan ahead, where is the first layer surface gonna be, could the layer adhesion be a weakspot in the chosen orientation, how can I orient the part for low friction, prevent overhangs to keep it support-free, and so on. All of that is extra effort for me, and lower effort for you during printing. And that is just the design-phase, with lots of additional work to prepare assembly instructions and proper documentation.

Why?

Because I care.

Because this is **DeSteiner**, your *rocksolid designer* for 3d-printing oriented designs.

On top of that, this thing is free to download, which does not mean that it has no value for you... Do you like my free creation and would like to show me your gratitude?

Leave me a tip/donation, it always makes my day and keeps new projects coming for you to download !

You can do so via paypal:

→ www.paypal.com/paypalme/tosteiner

Already sent me a donation?

Then cheers to you, my friend! ;)

Thanks for reading till the end, and have fun with your printed M1887!

Tobias ~DeSteiner

